

CSE 1321

Module 0

Algorithms and Abstractions



Goals

- Learn **Algorithmic Thinking**.
- Understand **what an algorithm is** and how to write an algorithm.
- Describe the concept of **Abstraction**.
- Describe the **relationship** between algorithms and programming

Topics

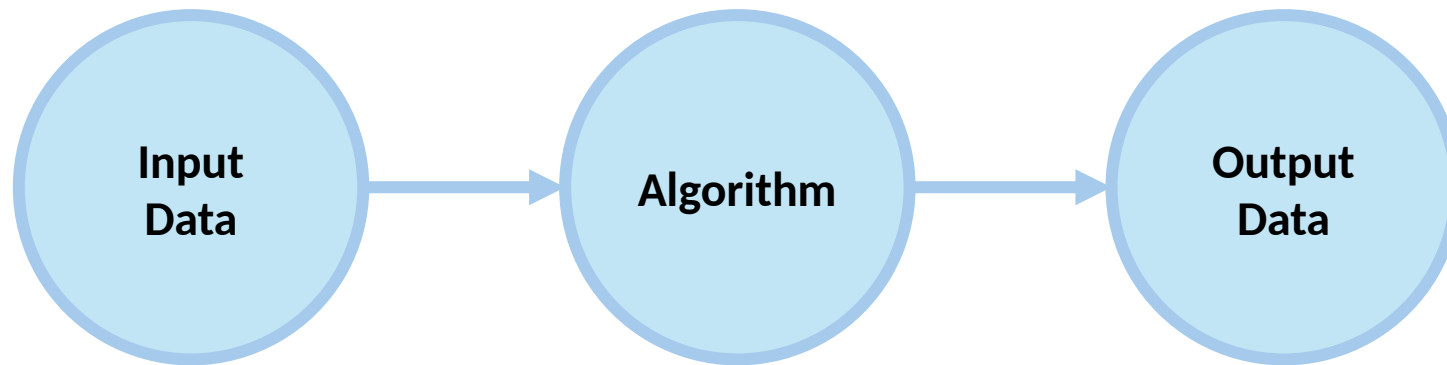
- Algorithms
 - Definition
 - Properties of Good Algorithms
 - Describing Algorithms
 - Examples
 - Components of Algorithms
- What is programming?

What is an Algorithm?

- A set of logical steps to accomplish a specific task.
- A set of directions/instructions.
- You use Algorithms everyday, even right now.

Algorithms in Programming

- An algorithm in programming contain:
 - Data
 - Instructions that process data



Algorithm Example: Chocolate Chip Cookies Recipe

- Ingredients:

- 2 1/4 cups flour
- 1 tsp baking soda
- 3/4 cup brown sugar
- 3/4 cup granulated sugar
- 12oz. semi-sweet chocolate chips
- 1 tsp salt
- 2 eggs
- 1 tsp vanilla ext.
- 1 cup soft butter

- Instructions:

- Preheat oven to 375°
- Combine flour, salt, and baking soda in a bowl. Set mixture aside
- Combine sugar, butter, and vanilla. Beat until creamy.
- Add eggs and beat.
- Add dry mixture and mix well.
- Stir in chocolate chips
- Drop mixture by teaspoons onto ungreased cookie sheet
- Bake 8 to 10 minutes.

Where or what is the input data, algorithm, and output data?

Abstraction – The level of detail

- Random House Dictionary of the English Language defines as:
 - A general idea or term.
 - An impractical idea; visionary and unrealistic.
 - General characteristics apart from concrete realities, specific object or actual instances.
- In Computing:
 - Abstraction refers to the logical grouping of concepts or objects.
 - We want to hide the details.
 - We like to think at a higher level of abstraction (no details).
 - Too many details overwhelm us!

Abstraction Activity

- Someone give me instructions to turn off the lights

What is wrong with this algorithm?

- From the back of a shampoo bottle:
 - Directions:
 - Wet hair
 - Apply a small amount of shampoo
 - Lather
 - Rinse
 - Repeat
- Just because we have an algorithm, does not mean it is a good one.

Properties of a good algorithm

- Good algorithms are:
 - Precise
 - Unambiguous
 - Complete
 - Correct
 - Simple
 - Contain levels of abstraction

Describing Algorithm

- Algorithms can be created using:
 - Natural language (e.g., English)
 - Pictures or flowcharts
 - A specific programming language
- For this course, we are going to use the programming language Python to implement algorithms into computer programs.

Natural Language Algorithm

1. START by make a list of courses you want to register for, in order of priority.
2. Number of hours = 0.
3. Choose highest priority class on list.
4. IF the chosen class is not full and its class time does not conflict with classes already scheduled, THEN register for the class:
 - 4.a. Add the class to the schedule.
 - 4.b. Add the class hours to the number of hours scheduled.
5. Cross that class off of your list.
6. Repeat steps 3 through 5 until the number of hours scheduled is ≥ 15 , or until all classes have been crossed.
7. END ALGORITHM.

Components of a programming algorithm

- Any computing algorithm will have at most five kinds of components:
 - **Data structures** to hold and manage data.
 - **Instructions** to change data value.
 - **Conditional** expressions to make decisions
 - **Control** structures to act on decisions
 - **Modules** to make the algorithm manageable by abstraction

What is programming?

- Programming is developing a set of instructions for the computer to act on.
- We develop an algorithm and we implement the algorithm in a programming language for the computer to execute.

How do I start programming? 1 of 2

- Programming requires two skills:
 - Algorithmic thinking
 - Knowledge of programming language syntax
- The syntax is the easiest of them both.
- Developing algorithmic thinking requires practice.
- To start programming:
 - Start by developing a good algorithm
 - Then, convert it into a language-specific syntax

How do I start programming? 2 of 2

- Think about the solution first, then write out the code.
- Not using a computer to write and run your program forces you to mentally execute your program and debug it.
- Too many programmers try to program by brute force and twiddling in an IDE. This is the wrong approach.

Summary

- Programming starts with algorithm development.
- An algorithm is just a set of instructions.
- Abstraction is just the level of detail.
- A good algorithm exhibits precision, clarity, completeness, correctness, and simplicity.
- An algorithm can be described using a natural language, pictures and diagrams, and a specific programming language.